MAS Project Guidelines & Grading Criteria

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Objectives

You will analyze a real dataset, draw statistical conclusions, and then prove your conclusions are worth believing. That last part is vital for the whole project. As for an example scenario, running a t-test takes ten seconds. Showing that the test's assumptions actually hold on your data, that the result survives resampling, and that the effect is large enough for anyone to care — that takes the rest of the semester. A submission that reports p-values without evidence about the assumptions behind them cannot pass, no matter how many methods it applies.

Work in teams of 6. Pick your own dataset and research questions, subject to the rules in the ethics section below.

Deliverables

1. A written report, 3 to 4 pages, not counting appendices.
2. A code repository that runs end to end from raw data to every figure and every number in the report.
3. A 12-minute presentation with 5 minutes of questions.

Grading Criteria

	Criterion	**Weight**
1	Research question and decision	10
2	Data and exploratory analysis	10
3	Distributional modeling evidence	15
4	Estimation and uncertainty	15
5	Hypothesis testing evidence	15
6	Statistical modeling and validation	15
7	Ethics and data handling	10
8	Documentation and reproducibility	10

Each criterion below tells you what to show and how to show it. Treat this section as the instructions.

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1. Research question and decision (10 marks)

Name one specific question and one specific person who would act on the answer. Not "**insights**." Not "**understanding the data**."

A hospital administrator deciding whether to add prenatal-visit reminders is a person and a decision. A landlord deciding whether to reprice units above the district median is a person and a decision. "**Exploring the relationship between the variables**" is neither, and it will cost you the marks.

Write one paragraph covering who uses the answer, what they do next, and what happens if the conclusion is wrong. Everything else in your project follows from this paragraph, so write it first and check it still holds at the end.

2. Data and exploratory analysis (10 marks)

Say where the data came from and what you did to it before analysis. Every consequential decision — dropped rows, missing values, transformations, outliers — goes in the report with a reason. Then explore the data thoroughly enough that a reader understands its shape before your analysis begins. Every figure you include must earn its place; a figure nobody interprets is decoration, and decoration earns nothing.

3. Distributional modeling evidence (15 marks)

Model at least one variable with a suitable probability distribution. Justify the choice from how the data is generated, not from the histogram shape alone, and provide evidence that the fit is adequate — evidence a skeptical reader could check, not an impression. Where the fit breaks, say so and say what it means for the conclusions that depend on it. "The data looked approximately Normal" earns nothing here.

4. Estimation and uncertainty (15 marks)

Every estimate you report comes with its uncertainty quantified and correctly interpreted. Choose the quantities your research question cares about, construct intervals for them, and show in some way of your choosing that those intervals can be trusted. Misinterpreting what an interval claims tells the grader the concept didn't land, and it will be graded that way.

5. Hypothesis testing evidence (15 marks)

Test hypotheses that matter to your question from criterion 1, stated before you look at the results. Assumptions are graded, not assumed. Whatever test you choose, show the evidence that it was valid for your data, and report effect sizes alongside p-values. Be transparent about everything you tried — a significant result found on the ninth attempt is not the same evidence as one found on the first, and pretending otherwise is the statistical version of hiding your failed runs.

6. Statistical modeling and validation (15 marks)

Fit the model your question needs. Any model from the course or real world is welcome. Choose it to fit the question, not to impress the grader; a well-diagnosed simple model beats a poorly understood complex one every time. Compare against a sensible baseline, run the

diagnostics your model type demands, validate out of sample, and interpret the outputs in units a decision-maker can act on. "X3 was significant" is not an interpretation.

7. Ethics and data handling (10 marks)

This covers how you got the data and how you analyzed it. It is graded on process, not on producing a philosophy essay.

Data collection. Say where the data came from and what its licence permits. Public repositories are fine. Scraped data usually isn't — check the terms before you scrape, not after. If your data includes anything about identifiable people, remove direct identifiers and say in the report what you removed.

Analytical honesty. Report every hypothesis you tested, not just the ones that came out significant. If you transformed a variable or excluded observations, the report says what the result looked like before you did. Removing outliers to force normality is analysis fraud in miniature, and it's the specific failure your lecture on convergence warned you about.

Two or three paragraphs on this is plenty. What earns the marks is specificity about your own data and your own analytical choices, not a survey of research ethics.

8. Documentation and reproducibility (10 marks)

Someone else should be able to clone your repository, run one command, and get every figure and every reported number back. Pin your dependency versions. Fix your random seeds — the permutation test, the cross-validation splits, and any posterior sampling all use them — and say what they are. Write a short data card covering the source, the licence, the cleaning steps, and known limitations. If a number in the report can't be regenerated from the code, it doesn't count as evidence.

Cautions

Every year a few projects fail in the same ways, so here are some reasons in advance.

Someone spends six weeks on data cleaning and feature engineering and two days on inference. **The cleaning isn't graded on effort. The statistical reasoning is.**

Someone produces a beautiful pairplot and stops. No fitted distribution, no interval, no assumption check. That's criteria 3 through 6 gone — more than half the marks.

Someone runs the test five ways, gets $p = 0.04$ once, and reports only that run. Report all five. Selective reporting found in the repository history costs more than an insignificant result ever would.

Someone writes "the mean is higher in group A, which makes sense." That sentence is doing no work. Either you have an interval or a test that says the difference is real, or you have a feeling that it should be.

Someone deletes eleven outliers because the Q–Q plot looked better without them. Eleven observations is eleven pieces of information about your population. Read your own justification out loud before you put it in the report.

And someone interprets a 95% confidence interval as a 95% probability in the final presentation, in front of everyone, in week 15. Criterion 4 exists so that this is not you.